THE UNIVERSITY OF HONG KONG
FACULTY OF BUSINESS AND ECONOMICS

MSBA7002 Business Statistics

GENERAL INFORMATION

Instructors: Dr. Weichen Wang
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Dr. Zhanrui Cai
Assistant professor, Innovation and Information Management
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Office: KK 1336

Consultation times: Before/after class, or by appointment

Tutor: Yutao Deng ytdeng@connect.hku.hk

Pre-requisites:

Basic mathematical skills, especially a knowledge of algebra. Experiences with using statistical softwares. We focus on the use of R – a powerful statistical computing package. It is open source and free. General information about R is available through https://www.R-project.org, including installation instructions. A user-friendly interface of R called Rstudio can also be downloaded https://www.rstudio.com/products/rstudio/download/.

An Introduction to R: http://cran.r-project.org/doc/manuals/R-intro.pdf.

Quick R: http://www.statmethods.net.

Co-requisites: None

Mutually exclusive: N/A

Course Website: Moodle

Other important details:

Required textbook:

Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani, *An Introduction to Statistical Learning with Application in R*, Second Edition, 2021, Springer.

https://www.statlearning.com

Optional textbook:

Trever Hastie, Robert Tibshirani, Jerome Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, Second Edition, 2008, Springer.

http://statweb.stanford.edu/~tibs/ElemStatLearn/download.html

Peter Dalgaard, *Introductory Statistics with R*, Second Edition, 2008, Springer.

http://www.academia.dk/BiologiskAntropologi/Epidemiologi/PDF/Introductory_Statistics_with_R__2nd_ed.pdf

Garrett Grolemund, Hadley Wickham, *R for Data Science*, First Edition, 2017, O'Reilly.

http://r4ds.had.co.nz

COURSE DESCRIPTION
The modern business world is increasingly experiencing data tsunami, i.e. being flooded with large amount of big data with complex structure and high dimensionality. Decision makers have begun to recognize the importance of “turning data into value” and the need for effective business analytics. This course systematically introduces various statistical tools, and the ability to apply them in business situations. Hands-on experience will be gained through real data analysis. Disscussions and practices will be offered about how to ask the right question, prepare the right data, and perform the right analysis to assist decision making, as well as ethical issues of data analytics.

COURSE OBJECTIVES
1. Obtain solid understanding about common statistical methods in business situations 2. Formulate the right business problem and identify suitable analysis tools 3. Carry out the analysis using software tools 4. Present analysis results in business relevant language 5. Understand ethical issues of data analytics

PROGRAMME LEARNING OUTCOMES
PLO1: Acquisition and internalization of knowledge of the programme discipline PLO2: Application and integration of knowledge PLO3: Inculcating professionalism and leadership PLO4: Developing global outlook PLO5: Mastering communication skills

COURSE LEARNING OUTCOMES

Course Learning Outcomes	Aligned Programme Learning Outcomes
CLO1: Clearly identify and formulate the relevant business problem	PLO1
CLO2: Select and use effective methods to address the business problem	PLO2
CLO3: Use software tools to provide solution to the issue at hand	PLO2, 4
CLO4: Communicate the solution effectively	PLO3, 5

COURSE TEACHING AND LEARNING ACTIVITIES

Course Teaching and Learning Activities	Expected contact hour	Study Load (% of study)
T&L1. Interactive lectures	30	25%
T&L2. Assignments, exercises, tutorials	60	50%
T&L3. Self-study, group discussion	30	25%
Total	120	100%

Assessment Methods	Brief Description (Optional)	Weight	Aligned Course Learning Outcomes

A1. Participation A2. Assignments A3. Final exam	Attendance and discussions Effort and accuracy Effort and accuracy	20% 40% 40%	CLO1, 2, 3, 4 CLO1, 2, 3, 4 CLO1, 2, 3, 4
	Total	100%	

STANDARDS FOR ASSESSMENT

Course Grade Descriptors

A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • High participation in discussions • Present arguments that have an element of originality • Achieve a standard of excellent performance in the exams with very accurate computation and very good analytical and problem solving skills • Excellent performance in assignments
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way • Good participation in discussions • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the exams with accurate computation and good analytical and problem solving skills • Good performance in assignments
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Some participation in discussions • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the exams with reasonably accurate computation and acceptable analytical and problem solving skills • Acceptable performance in assignments
D+, D	• Demonstrate a minimum understanding of the concepts involved • Barely address questions as set • Minimal or no participation in discussions • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the exams with some errors in computation and barely adequate analytical and problem solving skills • Marginally acceptable performance in assignments
F	• Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Minimal or no participation in discussions • Present arguments poorly • Fail to meet a standard of passing the exams with major errors in computation and inadequate analytical and problem solving skills • Poorly performance in assignments

Assessment Rubrics for In-Class and Tutorial Participation

A+, A, A-	• High participation in discussions • Always attend the tutorials and in-class discussions • Demonstrate a strong understanding of all relevant knowledge • Handling questions professionally • Present arguments that have an element of originality • Respect others and follow the class rules (no chatting and do not use cell phone)
B+, B, B-	• Good participation in discussions • Often attend the tutorials and in-class discussions • Demonstrate a good understanding of all relevant knowledge • Handling questions in a logical way • Present arguments that go beyond the lecture and textbook • Respect others and follow the class rules (no chatting and do not use cell phone)

C+, C, C-	• Some participation in discussions • Sometimes attend the tutorials and in-class discussions • Demonstrate a basic understanding of the concepts involved • Fairly address questions as set • Present arguments in a well-structure manner • Respect others and follow the class rules (no chatting and do not use cell phone)
D+, D	• Minimal or no participation in discussions • Rarely attend the tutorials and in-class discussions • Demonstrate a minimum understanding of the concepts involved • Barely address questions as set • Present arguments in a marginally acceptable manner • Respect others and follow the class rules (no chatting and do not use cell phone)
F	• Minimal or no participation in discussions • Almost never attend the tutorials and in-class discussions • Demonstrate a poor understanding of the concepts involved • Unable or unwilling to handle questions • Present arguments poorly • Behave poorly in class (often chatting with others, using cell phones, or being late)
Assessment Rubrics for Assignments and the Exam	
A+, A, A-	• Demonstrate a strong understanding of all relevant knowledge • Present arguments that have an element of originality • Achieve a standard of excellent performance in the assessments with very accurate computation and very good analytical and problem solving skills
B+, B, B-	• Demonstrate a good understanding of all relevant knowledge • Present arguments that go beyond the lecture and textbook • Achieve a standard of good performance in the assessments with accurate computation and good analytical and problem solving skills
C+, C, C-	• Demonstrate a basic understanding of the concepts involved • Present arguments in a well-structure manner • Meet a standard of acceptable performance in the assessments with reasonably accurate computation and acceptable analytical and problem solving skills
D+, D	• Demonstrate a minimum understanding of the concepts involved • Present arguments in a marginally acceptable manner • Meet a standard of marginally acceptable performance in the assessments with some errors in computation and barely adequate analytical and problem solving skills
F	• Demonstrate a poor understanding of the concepts involved • Present arguments poorly • Fail to meet a standard of passing the assessments with major errors in computation and inadequate analytical and problem solving skills

COURSE CONTENT AND TENTATIVE TEACHING SCHEDULE

The first five lectures will be delivered by Dr. Zhanrui Cai, and the last five will be delivered by Dr. Weichen Wang.

The listed topics below are tentative and subject to change, under the discretion of the instructors.

Lecture 1: Overview
- Overview of Business Analytics
- Review of Linear Regression

Lecture 2-3: Model Selection
- Linear Model Selection and Regularization
- Resampling Methods and Cross-validation

Tutorial 1, tutorial time to be announced

Lecture 4-5: Classification
- Logistic Regression
- Multinomial Regression

Tutorial 2, tutorial time to be announced

Lecture 6-8: Supervised Learning
- Poisson Regression

- Linear Discriminant Analysis
- Support Vector Machine

Tutorial 3, tutorial time to be announced

Lecture 8-9: Unsupervised Learning

- Principal Component Analysis
- Clustering

Tutorial 4, tutorial time to be announced

Final Exam: Time and location to be announced.

REQUIRED/RECOMMENDED READINGS & ONLINE MATERIALS (e.g. journals, textbooks, website addresses etc.)

To be announced on Moodle.

MEANS/PROCESSES FOR STUDENT FEEDBACK ON COURSE

X Conducting mid-term survey in additional to SETL around the end of the semester

○ Online response via Moodle site

○ Others: ________________________ (please specify)

COURSE POLICY (e.g. plagiarism, academic honesty, attendance, etc.)

An orderly learning environment is extremely important for this course. Disruptive behaviors are absolutely unacceptable. Academic dishonesty includes cheating, plagiarism, unauthorized collaboration, falsifying academic records, and any act designed to avoid participating honestly in the learning process. Any such dishonesty will result in an F grade.

ADDITIONAL COURSE INFORMATION (e.g. e-learning platforms & materials, penalty for late assignments, etc.)

Assignments will be submitted in class before the lectures. No late assignment will be accepted.
Lecture notes and self-learning materials will be uploaded on Moodle.
Good questions and discussions on Moodle will be counted as class participation.